

Summary

Materials scientist with a Ph.D. in Chemistry and postdoctoral experience at Argonne National Laboratory. Experienced in polymer synthesis, electrochemical materials, polymer membranes, and spectroscopic characterization. Skilled in NMR, FTIR, Raman, EIS, CV, and ionic-conductivity measurements. Author of 13 peer-reviewed publications, including three first-author papers published in the *Journal of the American Chemical Society*, *Inorganic Chemistry*, and *ACS Applied Materials & Interfaces*.

Skills

- **Characterization:** multinuclear NMR (^1H , ^{13}C , ^{15}N , ^{19}F , ^{31}P ; variable-temperature; 2D); FTIR; Raman; UV-Vis; fluorescence spectroscopy; SAXS/GISAXS at the Advanced Photon Source; DLS; GPC
- **Electrochemical Methods:** Cyclic voltammetry (CV); electrochemical impedance spectroscopy (EIS); ionic-conductivity measurements; ion permselectivity measurements
- **Materials Synthesis:** Polymer membrane synthesis and fabrication; thin-film preparation; transition-metal complex synthesis; inert-atmosphere synthesis
- **Research & Data Tools:** Mnova; Origin; Python; scientific data analysis and visualization
- **Laboratory Operations:** SOP development; EHS protocol implementation; laboratory setup; GPC instrument specification evaluation and vendor coordination

Professional Experience

Postdoctoral Appointee – Argonne National Laboratory Jan 2025 – Present

- Synthesized polymer membranes with varied functional groups and developed GPC-monitored protocols to control molecular weight and improve batch-to-batch reproducibility.
- Delivered qualified membrane samples to industrial collaborators and evaluated ionic conductivity across batches using EIS.
- Supported SAXS/GISAXS measurements at the Advanced Photon Source to characterize membrane morphology.
- Established EHS workflows and laboratory infrastructure for membrane synthesis and contributed to a pending fuel-cell membrane patent.

Organic Chemist (Intern) – KLA May – Aug 2024

- Synthesized rhodamine-based photoswitchable dyes for inspection technology development and characterized optical properties using UV-Vis and fluorescence spectroscopy.

Graduate Teaching/Research Assistant – Michigan State University Sep 2018 – Dec 2024

- Investigated Ru-catalyzed ammonia oxidation using variable-temperature NMR, isotope labeling, and Raman spectroscopy.
- Identified a previously uncharacterized dinitrogen-bridged intermediate, leading to a first-author publication in the *Journal of the American Chemical Society*.
- Led multinuclear NMR characterization and mechanistic analysis in a multidisciplinary electrochemical research collaboration, resulting in a first-author publication in *Inorganic Chemistry*.
- Trained junior researchers and served as a teaching assistant for General and Organic Chemistry.

Education

Ph.D. in Chemistry, Michigan State University, USA	2018 – 2024
M.S. in Chemistry, National Sun Yat-Sen University, Taiwan	2012 – 2014
B.S. in Chemistry, Chung Yuan Christian University, Taiwan	2008 – 2012

Research Highlights

- 13 peer-reviewed publications, 3 first author (full list on [Google Scholar](#))
- 1 patent application pending
- Oral presentations at 249th ECS Meeting **2026** and ACS Fall **2023** national meetings